

FaceCluster: Interactive Photo Organization with Enhanced Face Recognition

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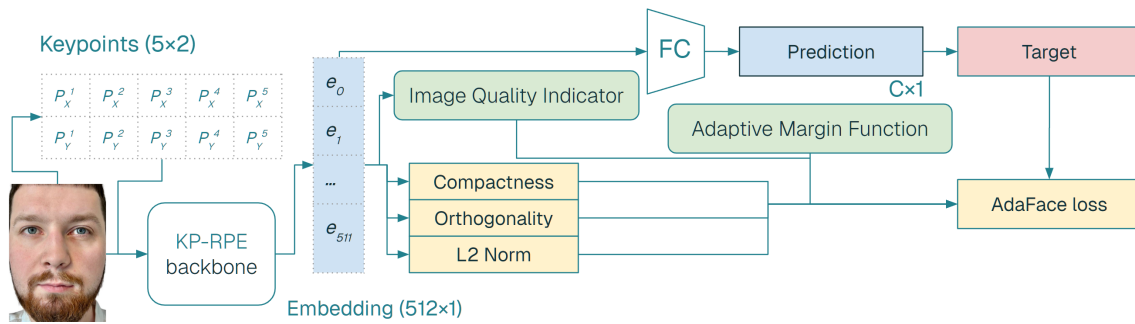


Figure 1: Architecture of the Statistical Embedding Regularization Approach.

Abstract

FaceCluster is an interactive photo management system that leverages our enhanced KP-RPE face recognition model with Embedding Statistical Regularization to organize personal photo collections automatically. Unlike existing cloud-based systems that raise privacy concerns, FaceCluster operates entirely locally while demonstrating high performance across multiple challenging benchmarks, including IJB-C (97.25% TAR@0.01%), TinyFace (74.14% Rank-1), and AgeDB (97.78% accuracy) when trained on the WebFace4M dataset. The demo showcases real-time face detection, clustering, and organization capabilities through an intuitive web interface, enabling users to effortlessly manage large photo collections with a single-command Docker deployment.

CCS Concepts

• **Computing methodologies** → **Biometrics; Matching; Object recognition.**

Keywords

Face Recognition, Embedding Regularization, Photos organization

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1 Introduction

Personal photo organization [10, 12, 14, 20, 25] remains challenging due to face recognition systems' limitations with pose variation, image quality differences, and diverse environmental conditions [3, 16, 21, 26]. While commercial solutions like Google Photos require cloud uploads and lack algorithmic transparency, research systems typically focus on benchmark performance rather than practical deployment [2, 4–6, 17–19].

In this paper, we introduce FaceCluster tool to address these limitations through three key innovations:

- (1) Enhanced Embedding Statistical Regularization applied to the KP-RPE [9] transformer backbone enforces compactness, orthogonality, and L2 constraints directly on face embeddings during training. This differs from traditional metric learning approaches by structuring the embedding space rather than just optimizing distances.

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- (2) Training-Free Deployment Architecture combining ONNX-optimized models with containerized deployment, eliminating the gap between research and practical applications.
- (3) Interactive Clustering Refinement using DBSCAN [23] optimized for unknown cluster numbers, particularly effective for organizing adult photo collections spanning decades.

Our approach builds upon the transformer-based KP-RPE backbone, which uses facial keypoints for spatial awareness, extending it with statistical regularization techniques. Unlike supervised approaches requiring domain-specific training data, our regularization operates during training to create better-structured embedding spaces without inference overhead.

2 Methodology

Our approach enhances the KP-RPE [9] face recognition backbone [4, 7, 15, 16] with AdaFace loss [8] for its quality-adaptive margin benefits. Our enhancements are applied during training to improve the resulting embeddings as shown in Fig. 1. We apply direct statistical constraints to the embeddings $\mathbf{e} \in \mathbb{R}^D$ produced by the backbone to improve their structure and discriminability. The combined regularization loss $\mathcal{L}_{\text{reg}}$ includes:

- **Compactness Loss**, initially used with triplet loss [22], encourages embeddings of the same identity to cluster tightly around their class center μ_c :

$$\mathcal{L}_{\text{compact}} = \frac{1}{|S_c|} \sum_{i \in S_c} \|\mathbf{e}_i - \mu_c\|_2^2 \quad (1)$$

where S_c is the set of samples belonging to class c , $\mathbf{e}_i$ is the embedding of sample i , and μ_c is the center of class c .

- **Orthogonality Loss** encourages orthogonality between different embeddings to improve feature disentanglement:

$$\mathcal{L}_{\text{ortho}} = \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1, j \neq i}^n (\mathbf{e}_i^T \mathbf{e}_j)^2 \quad (2)$$

where n is the batch size.

- **L2 Regularization** controls embedding magnitude:

$$\mathcal{L}_{L2} = \frac{1}{n} \sum_{i=1}^n \|\mathbf{e}_i\|_2 \quad (3)$$

The regularization components with weights 1.0 for $\mathcal{L}_{\text{compact}}$ (1), 0.5 for $\mathcal{L}_{\text{ortho}}$ (2), and 0.01 for $\mathcal{L}_{L2}$ (3) directly optimize embedding properties, going beyond traditional loss-based approaches to create more discriminative 512-dimensional representations. Table 1 presents a comparison with the baseline model when trained on the WebFace4M [30] dataset. To evaluate the effectiveness of our approach in enhancing face recognition performance across diverse scenarios, we conducted experiments incorporating embedding regularization on multiple benchmark datasets, including TinyFace [1], AgeDB [13], CFP-FP [24], IJB-C [11], IJB-B [27], CPLFW [28], and CALFW [29].

Table 1: Comparison to KP-RPE ViT-Small baseline trained on WebFace4M

Method	TinyFace Rank-1	AgeDB Acc.	CFP-FP Acc.	IJB-C TAR@0.01%	IJB-B TAR@0.01%	CPLFW Acc.	CALFW Acc.
KP-RPE [9]	69.88	-	96.60	94.20	-	-	-
Ours	74.14	97.78	98.94	97.25	95.75	95.33	96.08

3 System Architecture and Interactive Features

FaceCluster implements a microservices architecture with a FastAPI backend, a React frontend, and Redis caching. The enhanced KP-RPE model generates embeddings resulting in high clustering performance without additional inference costs.

3.1 Real-Time Processing Pipeline

Face Detection and Recognition. Users upload photos through a drag-and-drop interface, triggering automatic processing: an ensemble of face detection neural networks using autoannotator¹, enhanced KP-RPE embedding generation, and DBSCAN clustering.

Photo Collection Management. The demo illustrates robust photo-collection organization: FaceCluster reliably groups images of the same person from young adulthood to old age. Whereas traditional systems falter with appearance changes, our model preserves accuracy over decades, as confirmed on challenging benchmarks.

3.2 Demo Implementation and User Experience

The system utilizes GPU-accelerated ONNX models with automatic CPU fallback. FaceCluster demonstrates effortless deployment through Docker Compose, orchestrating a FastAPI backend, a React frontend, a Redis cache, and a SQLite database.

Interactive Demonstration Scenario. Attendees upload photo collections spanning multiple decades, witnessing real-time clustering of individuals across different life stages. The system correctly groups young adult and mature photos of the same person while separating different individuals, showcasing robustness across diverse conditions (see supplementary video²).

4 Technical Impact and Applications

FaceCluster provides intelligent, privacy-preserving photo organization by operating entirely locally. Its applications range from family archives to professional digital asset management. We employ an enhanced embedding regularization technique that delivers research-grade accuracy, enabling reliable person identification over decades. This demonstrates practical value beyond academic benchmarks for real-world, long-term photo management scenarios.

5 Conclusion

FaceCluster brings state-of-the-art face recognition to practical photo organization. Embedding Statistical Regularization augments KP-RPE, delivering top accuracy on diverse benchmarks without extra cost. One-command installation, a responsive web UI, and a scalable architecture support both personal and enterprise deployments.

¹<https://github.com/filonenkoa/autoannotator/>

²<https://youtu.be/Dd068BcjAkk>

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